

Final Assessment - Practical – Development/Programming Assessment

Marks Break Down:

Section A: Question 1: 15 Marks

Section A: Question 2: 15 Marks

Section B: Question 1: 15 Marks

Section B: Question 2: 15 Marks

GitHub Profile – End to End - Assignments : 40 Marks

Course: Fullstack AI – Batch – 9 / 10

You have to program/Develop : 2 question dataset from Section A and 2 question from section B

Theme of Paper:

Data science, machine learning (ML), and deep learning (DL) are revolutionizing the U.S. healthcare industry by improving diagnosis, treatment, and operational efficiency. These technologies enable healthcare organizations to leverage large volumes of data for smarter decision-making and better patient outcomes. As healthcare costs and demand continue to rise, AI-driven solutions are becoming critical for delivering high-quality and cost-effective care.

Importance in the U.S. Healthcare Market

- **Early Disease Detection**

ML and DL help identify diseases faster and more accurately. Early diagnosis improves treatment success rates.

- **Personalized Healthcare**
Data science supports customized treatment plans based on patient data. This leads to better outcomes and patient satisfaction.
- **Predictive Care**
AI models can predict health risks before conditions worsen. Early intervention reduces complications and costs.
- **Medical Imaging Enhancement**
Deep learning improves the analysis of X-rays, MRIs, and CT scans. This helps clinicians make accurate diagnoses.
- **Hospital Efficiency**
Analytics optimizes resources, staffing, and patient scheduling. This improves operational performance and care delivery.
- **Cost Reduction**
AI identifies inefficiencies and unnecessary procedures. This helps lower healthcare expenses across the industry.
- **Faster Drug Development**
ML accelerates research by analyzing large biomedical datasets. This speeds up the discovery of new treatments.
- **Remote Patient Monitoring**
AI can analyze data from wearable health devices. This enables proactive and continuous patient care.
- **Fraud Detection**
Data science helps detect suspicious claims and billing activities. This reduces financial losses for healthcare providers and insurers.
- **Clinical Decision Support**
AI-powered tools assist doctors with evidence-based recommendations. This improves accuracy and quality of care.

=====Dataset:

Dataset 1. [USA Hospitals - Hospitals across the USA]

<https://github.com/ShahzadSarwar10/Fullstack-AI-BOOTCAMP-B-10/tree/main/FinalAssessment/USA-Hospitals>

Dataset 2.[FDA AI Medical Devices: Authorizations & Recalls]

<https://github.com/ShahzadSarwar10/Fullstack-AI-BOOTCAMP-B-10/tree/main/FinalAssessment/FDA-AI-Medical-Devices-Authorizations-Recalls>

Dataset 3:[Oregon Hospital Data]

<https://github.com/ShahzadSarwar10/Fullstack-AI-BOOTCAMP-B-10/tree/main/FinalAssessment/Oregon-Hospital-Data>

Dataset 4:[Top 10 Healthcare Companies in the USA]

<https://github.com/ShahzadSarwar10/Fullstack-AI-BOOTCAMP-B-10/tree/main/FinalAssessment/Top-10-Healthcare-Companies-in-the-USA>

Dataset 5:[UCI Heart Disease Dataset]

<https://github.com/ShahzadSarwar10/Fullstack-AI-BOOTCAMP-B-10/tree/main/FinalAssessment/UCI-Heart-Disease-Dataset>

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Section A: Machine Learning - SK LEARN [Marks 15+15]

Need to apply – Numpy, Pandas, Seaborn

Apply: Classification, Regression, Clustering, Ensemble Learning

Question/Task - 1:

Select one of the above dataset and apply.

Question/Task 2:

Next, Select one of the above different dataset and apply.

Steps - TO Do: [Neat, Clean, well-documented code]

Loading in DataFrame, Perform “Descriptive Statistics” Max- Operations, Input data analysis based on seaborn, Performing core operation of SK Learn, Perform metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Need demonstration of all skills, learn in this course. End to end. For standard implementation, no marks , will be given. AI generated code or copy pasted code, will get only zero marks.

Section B: Deep Learning [Marks 15 + 15]

Apply end to end, Apply – Numpy, Pandas, SeaBorn

Apply Time SERIES – RNN , LSTM and GRU

Steps - TO Do:[Neat, Clean, well-documented code]

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Design Model with related layers, Performing core operation of Tensflow-Keras based deep learning, Perform Model metric analysis

[10 Marks for exceptional creative deep data analytical skills demonstration. For standard implementation – these marks will NOT be given.]

Question/Task - 3:

Select one of the above dataset and apply.

Question/Task 4:

Next, Select one of the above different dataset and apply.

Need demonstration of all skills, learn in this course. End to end. For standard implementation, no marks , will be given. AI generated code or copy pasted code, will get only zero marks.